

Rajapandiyan Arivazhagan

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PROFESSIONAL SUMMARY

Software Engineer with 9+ years of experience, including 7 years at Zoho Corporation (ManageEngine), specializing in Generative AI, Large Language Model (LLM) application development, and Python backend automation. Designed and deployed to production an enterprise AI support platform used by 200+ support technicians, combining Retrieval-Augmented Generation (RAG), multi-agent orchestration, and custom Model Context Protocol (MCP) servers that securely connect AI agents to SQL databases, internal APIs, and enterprise applications. Fine-tuned open-source LLMs on internal support data and operate self-hosted AI infrastructure with prompt injection and data-masking guardrails, backed by deep experience in Windows reverse engineering, patch automation, and large-scale web crawling.

TECHNICAL SKILLS

Programming & Backend: Python, Java, JavaScript, SQL, FastAPI, REST API Design & Integration, JSON/XML Parsing, Workflow Automation

Generative AI & LLMs: Large Language Models (LLMs), OpenAI API, Ollama, Prompt Engineering, LLM Fine-Tuning, Function/Tool Calling, Structured Outputs (JSON Schema), Agentic AI, Multi-Agent Systems, Model Context Protocol (MCP), MCP Server Development, Text-to-SQL (Natural Language to SQL), Hermes Agents, Open WebUI, Local/Self-Hosted LLM Deployment

RAG & Semantic Search: Retrieval-Augmented Generation (RAG), Document Ingestion, Intelligent Chunking, Embedding Models, Semantic Search, Reranking, Hybrid Retrieval, Context & Memory Management, Knowledge Base Indexing

AI/ML Frameworks & Safety: LangChain, LangGraph, Hugging Face Transformers, PyTorch, SQLAlchemy, Streamlit, Plotly, Computer Vision, Prompt Injection Mitigation, Sensitive Data Masking (PII, internal identifiers), Static Application Security Testing (Semgrep, Bandit), OWASP Top 10, CWE

Databases & Vector Databases: Microsoft SQL Server, SQLite, ChromaDB, FAISS, Pinecone, pgvector, Elasticsearch

DevOps, Crawling & Data: Docker, Git, Linux, Windows Server, Selenium, BeautifulSoup (bs4), Jsoup, JavaScript-Rendered Content Handling, Automated Data Pipelines

Enterprise & Endpoint Engineering: Windows Patch Management, Third-Party Application Patching, Software Reverse Engineering, Installer Analysis (EXE, MSI, MSIX), Silent Installation, Software Packaging, Deployment Automation, Detection Logic, CVE Analysis, Endpoint Management

PROFESSIONAL EXPERIENCE

Member Technical Staff — ManageEngine Endpoint Central, Zoho Corporation

2019 – Present

Technologies: Python, LangChain, LangGraph, Hugging Face Transformers, PyTorch, OpenAI API, Ollama, MCP, ChromaDB, Pinecone, Elasticsearch, FastAPI, Docker, Microsoft SQL Server, Selenium, BeautifulSoup, Jsoup, Open WebUI

Generative AI & LLM Engineering

- Architected and deployed to production an enterprise AI support automation platform that ingests customer tickets via Zoho Desk APIs, analyzes technical issues, retrieves institutional knowledge, and generates context-aware troubleshooting procedures — adopted by 200+ support technicians as part of their daily workflow.
- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline over internal documentation, knowledge base articles, engineering notes, historical support tickets, and solution repositories using intelligent chunking, embedding generation, and semantic indexing.
- Improved retrieval precision by implementing reranking and hybrid retrieval over vector search, surfacing more relevant enterprise context and reducing irrelevant retrievals in LLM-generated responses.
- Developed custom Model Context Protocol (MCP) servers enabling AI agents to securely access Microsoft SQL Server databases, file systems, internal REST APIs, enterprise applications, and external web resources through governed, tool-based interfaces.
- Designed multi-agent architectures using LangChain and LangGraph in which specialized agents collaborate to interpret ticket context, query internal systems, retrieve documentation, reason over historical cases, and recommend validated solutions.
- Fine-tuned open-source LLMs on curated internal support data — customer queries, resolved answers, and product-specific troubleshooting logic — to improve domain accuracy and reduce generic model responses.
- Enforced structured outputs with JSON schema validation and function/tool calling, producing reliably parseable agent responses for downstream automation.
- Engineered AI guardrails including prompt injection mitigation and automated masking of sensitive data — mobile numbers, customer identifiers, and internal confidential information — before payloads reach language models.

- Deployed and operated self-hosted LLM infrastructure with Ollama, Open WebUI, and Docker, keeping enterprise data fully on-premises, and exposed AI capabilities as FastAPI REST services for integration with internal tools.

Software Intelligence, Web Crawling & Automation

- Built and maintained enterprise-scale web crawlers using Python, Selenium, BeautifulSoup, and Jsoup to continuously harvest software intelligence — versions, release notes, download URLs, patch metadata, security advisories, CVE data, and lifecycle information — across a large portfolio of software vendors.
- Engineered resilient crawlers handling JavaScript-rendered pages, authenticated sessions, dynamic content, pagination, and anti-scraping protections, sustaining reliable extraction despite frequent vendor site changes.
- Automated continuous vendor release monitoring to detect new versions and security patches immediately, and built Python automation tools, database utilities, and reporting systems that replaced manual engineering workflows.

Software Reverse Engineering & Deployment Automation

- Reverse engineered Windows installers (EXE, MSI, MSIX, bootstrap and custom setup packages) to determine silent install/uninstall switches, installer parameters, exit codes, prerequisites, registry modifications, file system changes, services, and package dependencies.
- Conducted installer analysis through process and registry monitoring, file system diffing, network traffic inspection, and runtime behavior analysis to uncover undocumented installation parameters, hidden APIs, and internal vendor endpoints.
- Built custom deployment workflows for applications lacking official silent installation support, plus installation validation, detection logic, post-install verification, and rollback handling for reliable unattended deployment across enterprise Windows endpoints.
- Resolved patch deployment failures through root cause analysis of installer behavior, environment conflicts, and Windows update supersedence.

Data Analyst — MicroPrecision

2017 – 2019

Technologies: Python (pandas, NumPy), SQL, Microsoft Excel, Data Cleaning, Exploratory Data Analysis, Reporting Automation, Data Visualization

- Automated recurring data collection, cleaning, and reporting workflows using Python and SQL, replacing manual spreadsheet processes and freeing analyst time for higher-value investigation.
- Wrote SQL queries to extract, join, and aggregate operational and production data from relational databases, delivering consistent datasets for business and engineering reporting.
- Built Python data pipelines using pandas and NumPy to transform, reconcile, and validate datasets, improving data consistency and reducing manual errors in downstream reports.
- Performed exploratory data analysis (EDA) on production and quality data to identify trends, outliers, and process anomalies, surfacing issues that were not visible in standard reports.
- Developed recurring operational dashboards and management reports in Excel and Python, translating raw process data into metrics that engineering and operations teams used for decision making.
- Implemented automated data quality and validation checks across reporting workflows, improving reliability of figures used in operational reviews.

KEY PROJECTS

AI-Powered Support Automation Platform — Python, LangChain, LangGraph, MCP, RAG, ChromaDB, Ollama, OpenAI API, FastAPI, Docker
Production multi-agent AI assistant used by 200+ support technicians. Retrieves tickets via Zoho Desk APIs, performs semantic retrieval with reranking across enterprise knowledge sources, queries SQL databases and internal APIs through custom MCP servers, and generates context-aware troubleshooting steps grounded in historical cases. Includes fine-tuned domain models, structured JSON outputs, prompt injection guardrails, and sensitive data masking on self-hosted infrastructure.

QueryMind (GenAI + Text-to-SQL) — Python, LangChain, SQLAlchemy, RAG, LLMs, Streamlit, Plotly

Intelligent data analytics assistant that converts natural-language questions into SQL queries using LLMs and executes them against relational databases. Combines LangChain, SQLAlchemy, and RAG for schema-aware query generation and contextual responses, with interactive Streamlit and Plotly visualizations enabling non-technical users to explore business data without writing SQL.

VulnGuard AI (LLM + Cybersecurity) — Python, FastAPI, LangChain, FAISS, Streamlit, Semgrep, Bandit, RAG, LLMs

AI-powered source code vulnerability analyzer combining LLM reasoning, static analysis, and Retrieval-Augmented Generation (RAG) to detect SQL injection, XSS, command injection, insecure authentication, hardcoded credentials, and unsafe dependencies. Integrates Semgrep and Bandit findings with LLM-based reasoning to explain vulnerabilities, assess severity, and generate secure remediation suggestions, grounded in OWASP and CWE knowledge indexed in a FAISS vector store for context-aware security recommendations.

EDUCATION

Bachelor of Engineering (B.E.), Mechanical Engineering — Kamaraj College of Engineering and Technology, 2017

Languages: English, Tamil